Title: Personalized Learning Path Prediction from Student Interaction Patterns

Abstract

Student interaction with online learning platforms provides valuable information about their learning behavior, progress, strengths, weaknesses, and engagement. Traditional learning systems generally provide the same sequence of learning materials to all students and often fail to adapt to individual learning patterns. This project proposes a **Personalized Learning Path Prediction from Student Interaction Patterns** framework that uses machine learning techniques to analyze student interactions and predict suitable next learning activities or learning paths.

The proposed system uses student interaction data such as **quiz attempts, assessment scores, time spent on learning materials, number of completed activities, content access frequency, previous performance, and learning activity sequences**. The data is preprocessed to handle missing values, normalize numerical features, and construct meaningful behavioral features. Machine learning models such as **Random Forest, XGBoost, or Logistic Regression** can then be trained to predict the student's most suitable next learning module or learning-path category based on their historical interaction patterns.

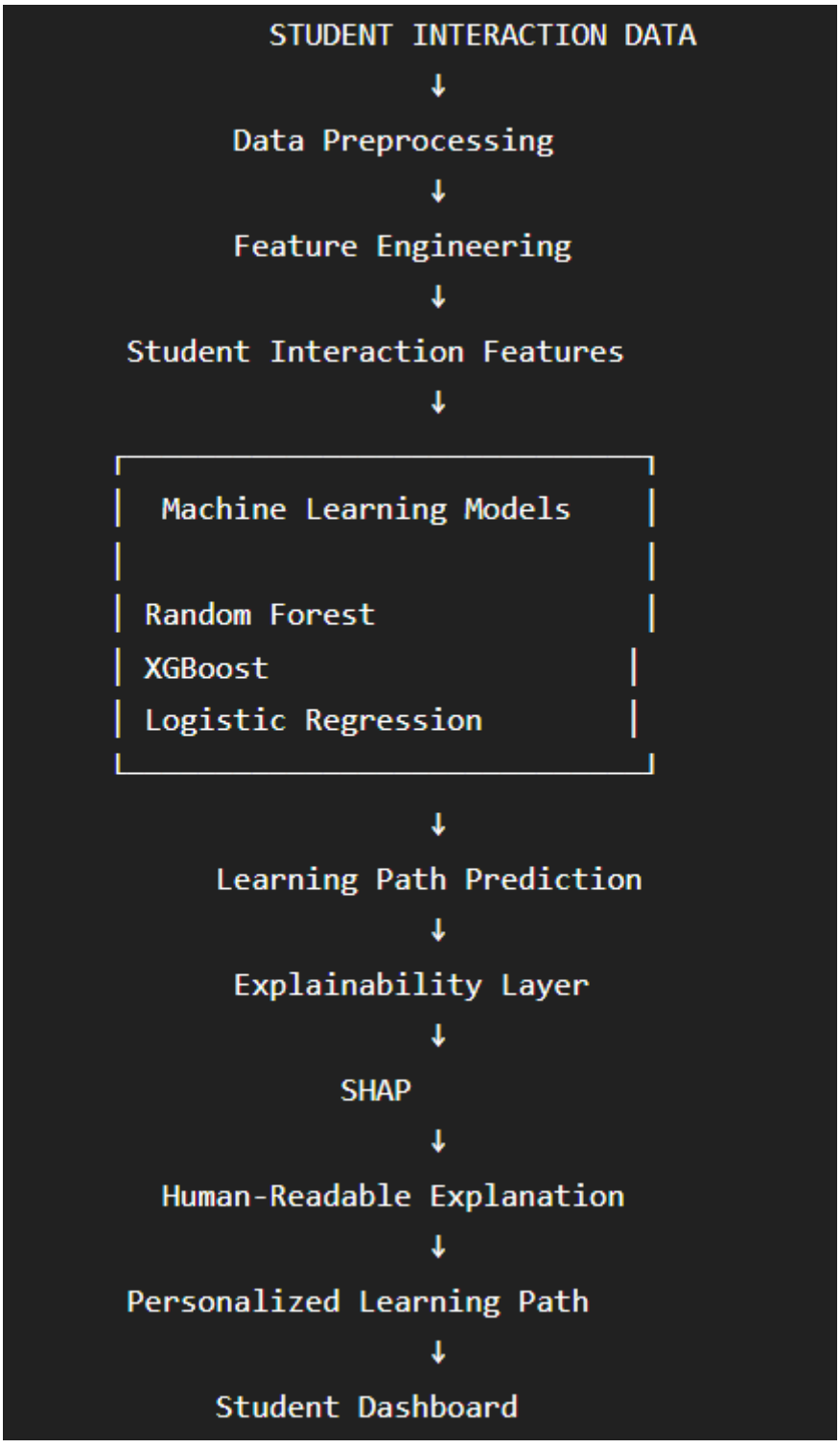
To improve the interpretability of the predictions, **SHAP-based explainability** can be incorporated to identify the student interaction features that contribute most significantly to a predicted learning path. For example, the system can determine whether low quiz performance, repeated attempts, reduced activity, or higher engagement with a particular topic influenced the recommendation. This allows the predicted learning path to be understandable rather than simply presenting a recommendation without justification.

The proposed system can further analyze student interaction patterns to identify **high-performing learners, students requiring additional support, frequently struggled topics, and changes in learning engagement**. The performance of the prediction models can be evaluated using **accuracy, precision, recall, macro-F1 score, confusion matrix, and feature importance**, depending on the selected prediction task. The resulting system aims to provide personalized and interpretable learning-path recommendations that can help students follow more suitable learning sequences and assist educators in identifying areas where additional academic support may be required.

Datasets

Dataset	URL link
EdNet	https://github.com/riiid/ednet

Approach:



Project Group Members

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